

Manpreet Singh

manpreet.s.cse@gmail.com | [linkedin.com/in/manpreet-singh-cse](https://www.linkedin.com/in/manpreet-singh-cse) | github.com/MSingh-CSE

Experience

SDE - AI | Ross Video

Feb 2026 – Present

- Exploring AI capabilities for next-generation broadcast and live production systems, developing agentic prototypes and intelligent tooling that integrate with video production infrastructure.
- Working on agentic AI systems, multi-modal pipelines, and MCP servers to assist operators and automate complex production workflows.

ML Research Engineer | Hypermatrix Labs, Dalhousie Uni.

May 2023 – Dec 2024

- Built automated parallelized pipelines for measuring the impact of quantization on neurons and internal representation of LLMs, providing detailed insights on **the efficacy of quantization in real-world production systems**.
- Designed embedding analysis workflows using manifold learning to detect model biases.

ML Engineer | TCS

Jul 2019 – Oct 2022

- Developed **Natural Language-to-SQL system (GPT-3)**, achieving 92% accuracy on 500+ daily queries.
- Deployed **Rasa chatbot** handling 1,000+ queries/day, improving developer productivity by 25%.
- Created **2D-to-3D CV pipeline** to generate 3D house model from 2D blueprint image, reducing visualization time by 20%.
- Collaborated with cross-functional teams to deploy ML services via Docker, Kubernetes, and REST APIs in production.

Education

MSc, Computer Science (Thesis based) | Dalhousie University

Jan 2023 – Dec 2024

- GPA (Graduate Courses):** 3.9 / 4.3 | **Relevant Coursework:** Machine Learning, Deep Learning, NLP, ML for Software Engineering

Bachelor of Computer Applications | Guru Nanak Dev University

May 2016 – May 2019

Publications

Manpreet Singh and Hassan Sajjad. "[Interpreting the Effects of Quantization on LLMs](#)." IJCNLP-AACL 2025, Mumbai, India. pp. 2267–2281.

Projects

Interpreting the effects of Quantization on LLMs | Python, PyTorch, NLP, Quantization, Prompt Engineering

- Created automated parallelized pipeline to perform analysis on neurons of 5 open-source LLMs measuring the impact of quantization on internal representations and behavior of LLMs.

RAG-Based Retrieval System for Open Canada Government Data | Python, LangChain, Qdrant, vLLM, Rest APIs

- Developed a RAG pipeline leveraging open-source models with vLLM, LangChain, and Qdrant to efficiently query and retrieve data from Canada's Open Government portal.

Natural Language to SQL Query | Python, SQL Integration, Rest APIs, LLMs

- Engineered a GPT-3 powered natural-language query interface that translates user prompts into SQL for live custom DBs and executes them through Python and REST API services.

Mythology Stories Generator | Python, Fine-tuning, Transformers

- Fine-tuned LLM to generate or continue plot of stories helping authors to explore possible continuations for stories.

Certifications

[Databricks Fundamentals](#) | [LLMOps \[DeepLearning.AI\]](#) | [Nvidia Deep Learning](#)

Key Skills

- Programming & Data:** Python (primary), SQL, Java, JavaScript, C
- ML & AI:** PyTorch, TensorFlow, Transformers, RAG, LLMOps, RLHF, Quantization, Finetuning, Explainable AI, Agentic AI Workflows & Architectures, Context Engineering, Prompt Engineering, Automated Pipelines, MCP
- MLOps & Production:** Kubeflow, Vertex AI, MLflow, Docker, Kubernetes, REST APIs, CI/CD, Cloud ML (GCP)
- Distributed & HPC:** Slurm, Multicore Optimization, Cluster Management

Extracurricular Achievements

Presented **Neural Style Transfer** project to India's Education Minister (TCS internal showcase).

Ranked in **Top 3 of 1000** participants in a 20-week AI/ML training program at TCS.